PROJECT ARES

AI-Adaptive Lunar Habitat Design

Psychological Wellbeing Through Intelligent Environments

8-Week Team Project

Teams of 3–4

200 Points Total

Psychology · Engineering · AI · Ethics

A student design challenge in space psychology and AI systems



The problem

Structural engineering on the Moon is largely solved. Crew psychology is not.

18

months

Mission duration — same as longest ISS stays

6

crew

People sharing a space smaller than a house

#1

risk

Behavioural health ranked top mission risk by NASA

0

outside

No parks, no fresh air, no spontaneous human contact

Your mission

Design the psychological support architecture for a six-person lunar habitat on an 18-month mission. Your system must use AI, biosensors, voice analysis, and facial recognition to detect and respond to crew psychological states — and adapt the living environment in real time.



Detect

Biosensors, voice prosody, facial recognition, and behavioural tracking feed continuous data to the AI



Analyse

AI cross-references signal streams, triages severity, and builds a longitudinal baseline for each crew member



Respond

The habitat adapts — lighting, audio, temperature, scent, schedules, and social nudges — automatically and proportionally

Six conditions to address

Your system must detect and respond to all six.

Anxiety

Acute stress from confinement, pressure, or safety fears

Depression

Sustained low mood and withdrawal building over days

Interpersonal conflict

Crew tension, avoidance, and communication breakdown

Cognitive fatigue

Declining task performance from sustained high-stakes work

Autonomy deprivation

Loss of personal agency in a tightly scheduled environment

Sensory monotony

Perceptual habituation and boredom from unchanging surroundings

Sensor architecture

Phase 1 (Weeks 1–2): Research which sensors detect which conditions

HRV
Heart rate variability
Anxiety · stress · fatigue

GSR
Galvanic skin response
Acute anxiety · arousal

Sleep
Sleep architecture tracker
Depression · fatigue · circadian

Voice
Voice prosody analysis
Mood · conflict · fatigue

FACS
Facial action coding
Emotion · depression · conflict

Motion
Movement / proximity
Isolation · conflict · autonomy

Cortisol
Cortisol wearable
Chronic stress · depression

Air
CO_2 / air quality
Cognitive fatigue · irritability



Key design question

Every sensor added to a lunar habitat costs mass and power. Justify why each sensor is worth its weight. What happens when one fails? How does your system gracefully degrade?

Habitat spatial design

Phase 2 (Weeks 3–4): Design zones for psychological purpose, not just function

Communal zone

Shared meals, conversation, group activities. Warm lighting to draw crew together.

Private retreat

Acoustically shielded. Each crew member's personal space for rest and decompression.

Work & mission bay

Cool 6500K lighting, structured airflow. Optimised for sustained cognitive performance.

Exercise bay

Energising light and audio auto-activate. Screens show Earth views. Movement is medicine.

Nature / growth bay

Plant growth area. Green light spectra, petrichor scent. Behavioural activation for depression.

Transition corridor

Lighting and temperature shift between zones create psychological cues for mode-switching.

Deliverable: Annotated floor plans for two configurations — nominal operations and conflict protocol

Room Design: Communal vs. Private Space

ENGINEERING DESIGN PROCESS
Define → Specify → Design → Test → Evaluate

Room 1: Communal Hub

PURPOSE

Shared meals, group conversation, leisure, and mission debriefs. The social anchor of the habitat — where crew cohesion is built and maintained over 18 months.

MANAGING THIS SPACE

- **Scheduling:** AI coordinates shared meal windows; prevents forced overlap for crew in conflict
- **Ambient cues:** Warm 2800K lighting and low-level scent draw crew without compulsion
- **Conflict protocol:** Acoustic zones create separation; spatial rearrangement reduces forced proximity

Design task: Scaled floor plan with dimensions, sensor placements, furniture layout, and lighting/audio zones. Justify every element.

KEY QUESTIONS

› How do you make participation feel chosen, not compelled?
› What minimum floor area prevents claustrophobia for six people sharing one room?
› How does the layout change during a conflict protocol?
› Where are sensors placed, and which crew members know they are active?
› How does the AI draw people together without making anyone feel watched or managed?

Room 2: Private Retreat Pod

PURPOSE

Individual decompression, sleep, privacy, and personal agency. The one space each crew member controls completely — including whether the AI has any access at all.

THE KEY DISTINCTION

- **AI access:** No cameras inside. Environmental controls only on crew member request. The AI monitors the door, not the room.
- **Individual control:** Lighting, temperature, and audio set entirely by the occupant. No AI override.
- **Physical boundary:** Lockable. No crew member may enter without permission. Design the lock mechanism.

Design task: Scaled floor plan with ergonomic sleep space, personal storage, sensory controls, and the privacy boundary. Specify dimensions.

KEY QUESTIONS

› What minimum volume allows a human to sleep and decompress without feeling sealed in?
› How does the AI detect distress if it cannot see or hear inside the room?
› What personal items does each crew member pre-load into their pod before launch?
› How does acoustic isolation work without making the occupant feel cut off from the crew?
› Under what exact conditions may the CMO or AI enter or monitor this space?

AI response system design

Phase 3 (Weeks 5–6): Design complete response protocols for each condition

Mild / early	Auto — silent background adjustment. No crew notification.
Moderate	Active intervention + crew notification. Options offered.
Severe	Immediate escalation to Chief Medical Officer.

Environmental levers available to the AI

Lighting
CCT · lux · circadian cycle · blue-light filter

Audio
Binaural beats · soundscapes · acoustic zones

Thermal
Temperature · airflow · radiant warmth

Scent
Lavender · citrus · petrichor · ion levels

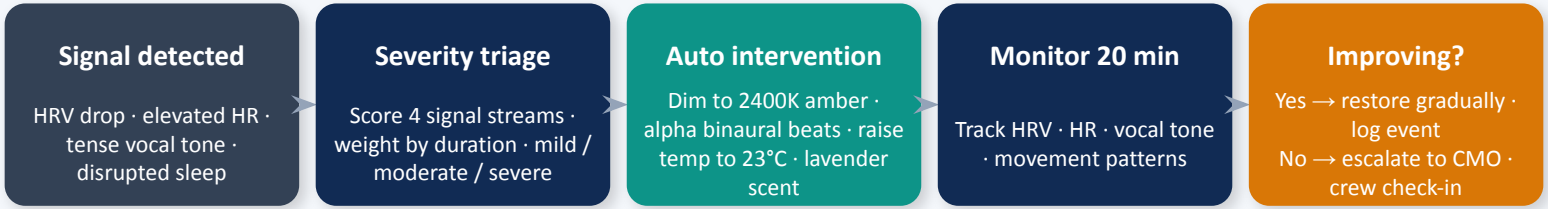
Routine
Schedule · activity prompts · breaks

Social
Earth calls · shared meals · AI companion

Visual
Home videos · nature scenes · Earth views

Example: Anxiety response flowchart

Students must produce flowcharts for at least 3 of the 6 conditions — here's the structure to follow



Speed matters for anxiety

Anxiety spikes fast. The AI must act within minutes. Monitor window is 20 min — not 20 hours.

Silent first, then escalate

Most calming interventions work better when the crew member isn't aware they're happening.

Never override consent

If a crew member explicitly refuses an intervention at any tier, the AI must stop and log.

Ethics & hard limits

Phase 4 (Weeks 7–8): The hardest design questions have no easy answers

Surveillance vs. privacy

The AI must watch continuously to help — but crew members live in the habitat 24/7. Where are the boundaries?

Autonomy vs. intervention

If a crew member refuses help, should the AI back off even if the crew's safety might be at risk?

Transparency vs. distraction

Should the AI explain what it's doing in real time — or would constant notifications themselves cause anxiety?

Individual vs. team

Crew member A's data reveals a risk. Can the AI act in a way that protects others without revealing A's private state?

Your ethics report must define hard limits: what the AI is NEVER permitted to do without human authorisation.

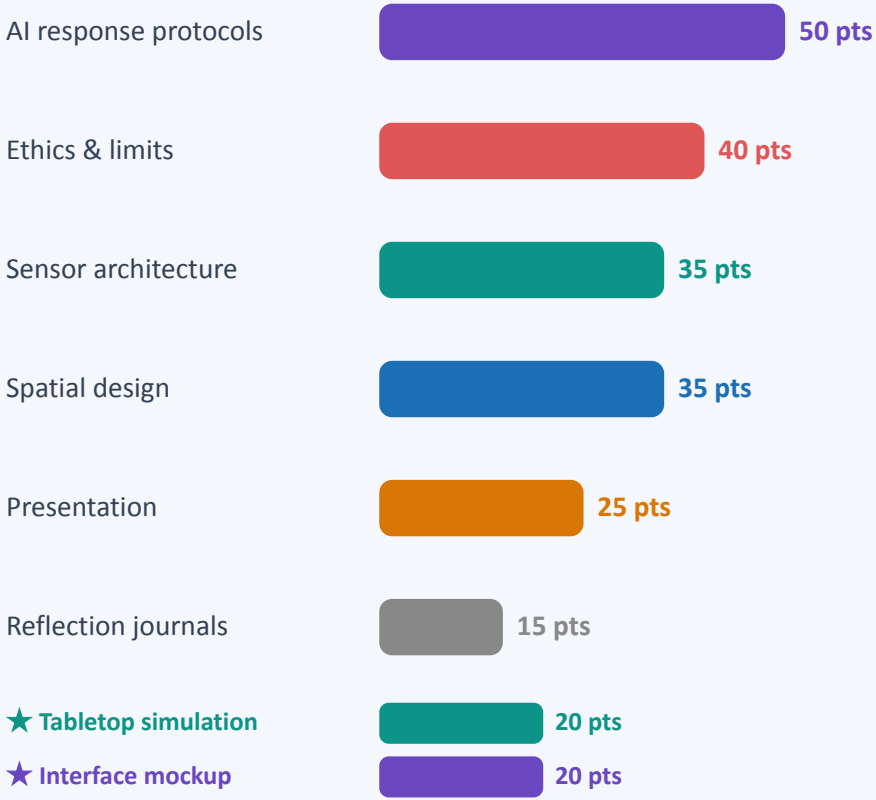
Deliverables & grading

Required deliverables

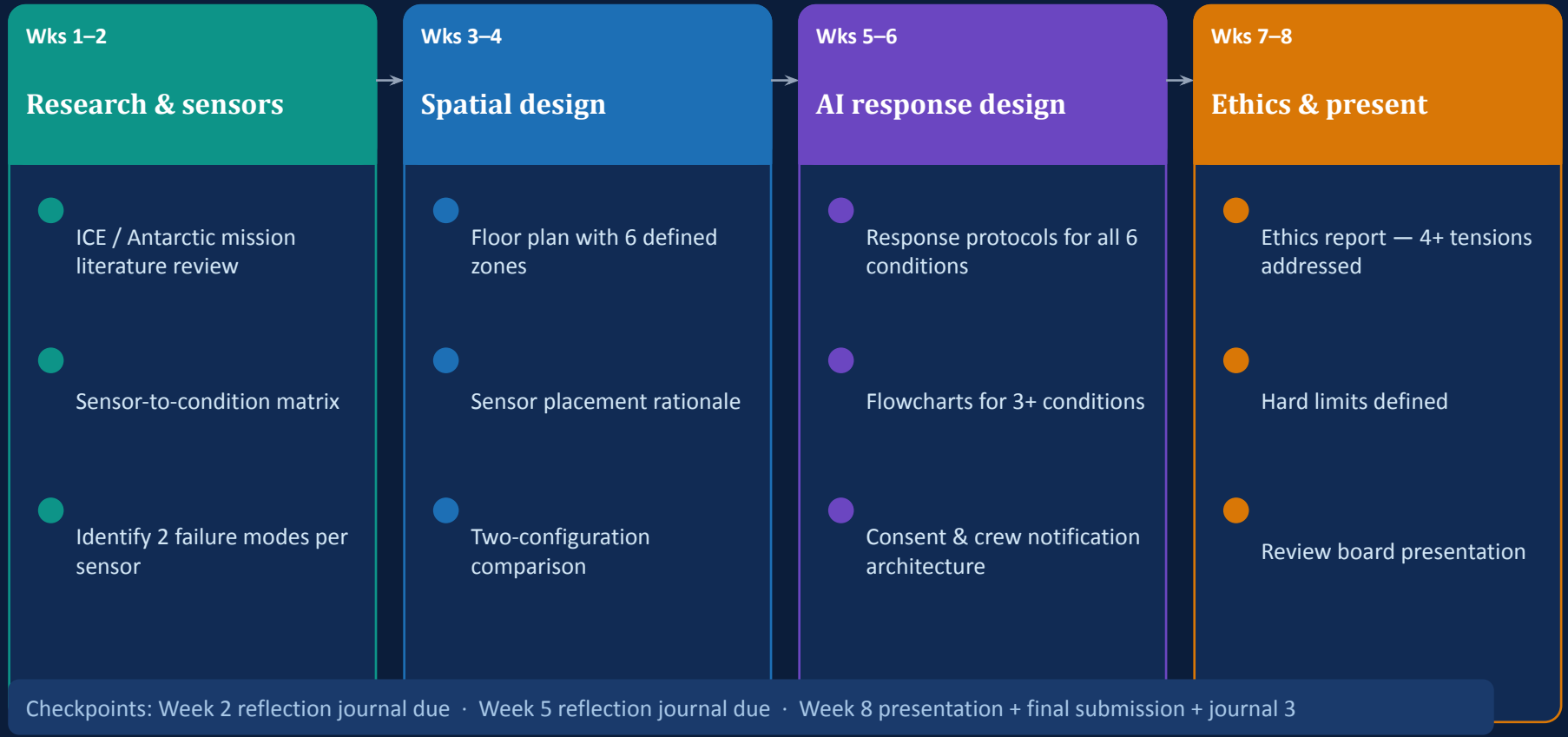
- Sensor-to-condition detection matrix
- Annotated floor plans — 2 configurations
- AI response flowcharts for 3+ conditions
- Ethics & limits report (800–1200 words)
- Agency review board presentation (15 min)
- Individual reflection journal — 3 entries
- ★ **Tabletop simulation — 20 pts**
- ★ **Interface mockup — 20 pts**

★ **HIGH-PRIORITY FOR DESIGN ENGINEERING:** Tabletop simulation and interface mockup are full rubric deliverables. In engineering design, testing

Points breakdown (240 total)



8-week timeline



Final review board

Week 8: Present to the agency



Defend every sensor choice on mass, power, and detection accuracy



Explain your AI severity triage logic under adversarial questioning



Walk the board through a live scenario: two conditions, simultaneously



Justify your hard limits — what the AI will never do, and why

15 minutes presentation · 10 minutes Q&A from the review panel

Good luck — the Moon is waiting.